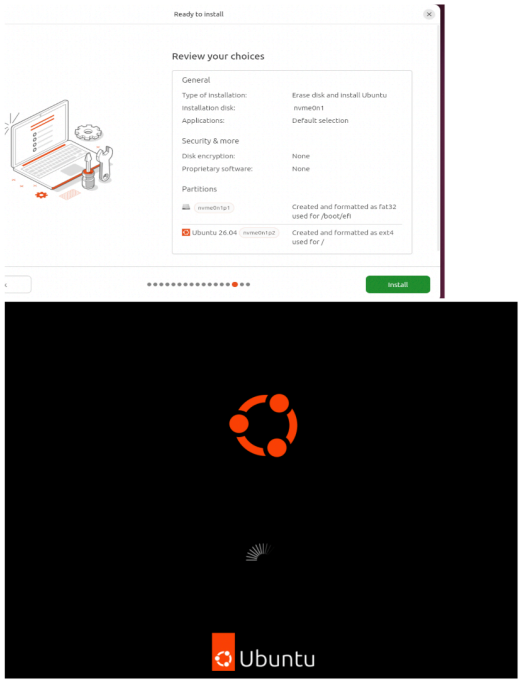
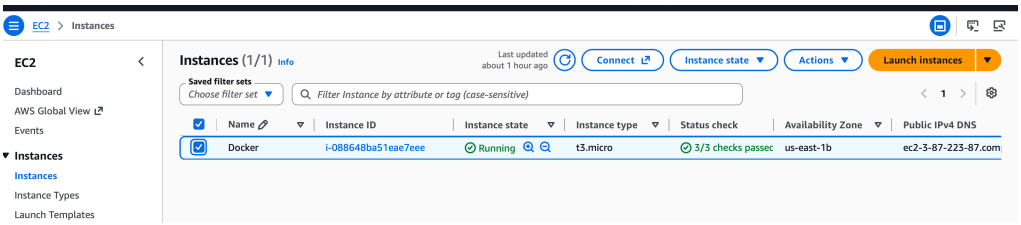
Docker-Task-1

1. Setup a virtual machine using Ubuntu image



2. Install docker on ec2.




```
ubuntu@ip-10-0-21-61:~$ sudo systemctl enable --now docker
ubuntu@ip-10-0-21-61:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-13 10:13:50 UTC; 1min 0s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2456 (dockerd)
      Tasks: 9
     Memory: 25.9M (peak: 26.8M)
        CPU: 320ms
     CGroup: /system.slice/docker.service
             └─2456 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

3. Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

```
ubuntu@ip-10-0-21-61:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
f7e1db57b58b: Pull complete
807ab930f3a0: Pull complete
ecc510c1e359: Pull complete
04bdbc7ea6b5: Pull complete
947d22046304: Pull complete
1efc53192927: Pull complete
8020170dc65f: Pull complete
c329293b1d90: Download complete
a1b85ab28c90: Download complete
Digest: sha256:f9ea18bfa4fad859e1ed38259d711da7ccad2c3516e875cec3351a57c859f571
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Docker-Task-1

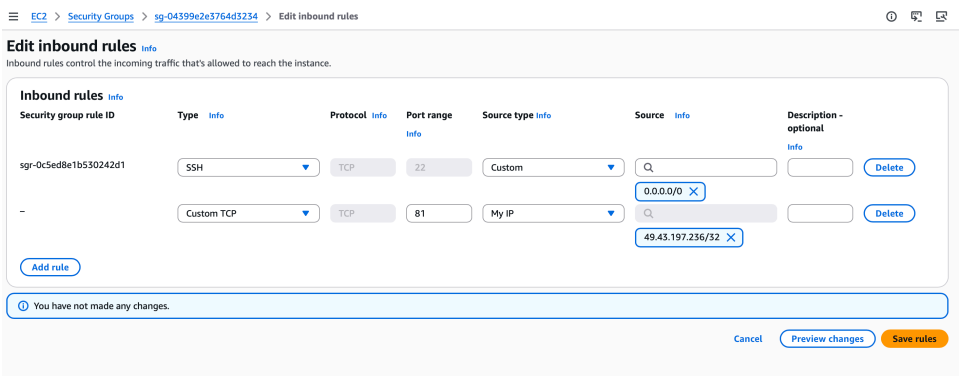
```
ubuntu@ip-10-0-21-61:~$ sudo docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
4e07a0f12b2c: Pull complete
06ad70e463aa: Pull complete
5dec61325458: Pull complete
f39241fdc590: Pull complete
e59a999a9c9e: Pull complete
130739ca07b1: Pull complete
0b2769356b03: Pull complete
89ac485ce7f9: Download complete
e0be5b59881d: Download complete
Digest: sha256:6974cddd46f09a10a0d4ffae6cd77f9122b3cf7de8f4b7c3305f35026accc81f
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
ubuntu@ip-10-0-21-61:~$
```

```
ubuntu@ip-10-0-21-61:~$ sudo docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
8f70d2bfe91a: Download complete
Digest: sha256:f144425ff09be612d6d9ad965196e9cdc23dae1f42110a8a11a3e9a8198759f7
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
ubuntu@ip-10-0-21-61:~$
```

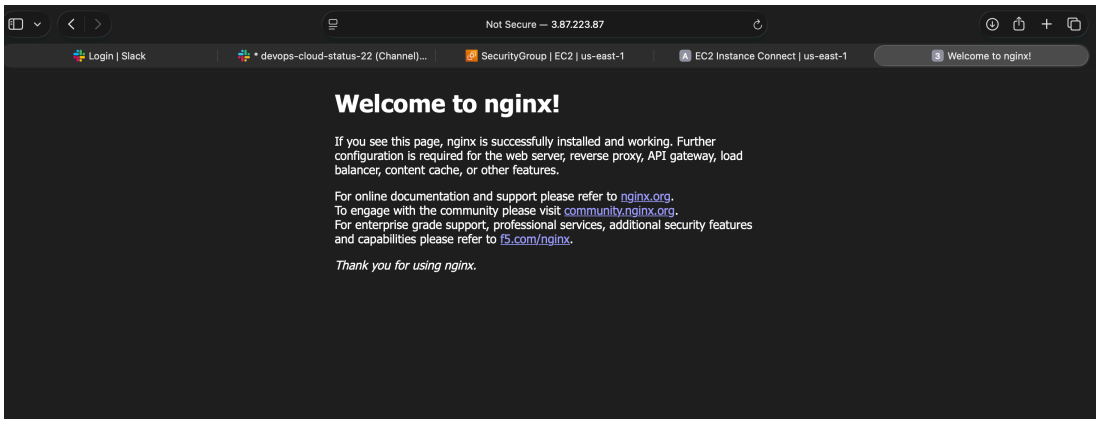
```
ubuntu@ip-10-0-21-61:~$ sudo docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
701ded2f9b33: Pull complete
a20a14499a3e: Pull complete
6eefb2f5d3e9: Pull complete
463a574793f8: Pull complete
408777b27aa9: Pull complete
810008851f3a: Pull complete
d89089c8a5a3: Pull complete
a91427fe0e3f: Pull complete
f9e31fab50e8: Pull complete
968c36a3b2c5: Pull complete
3dd4a99993ab: Pull complete
95bc7464aaea: Pull complete
44136fa355b3: Download complete
9613b4f6b0c7: Download complete
Digest: sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
ubuntu@ip-10-0-21-61:~$
```

```
ubuntu@ip-10-0-21-61:~$ sudo docker pull sonarqube:lts-community
lts-community: Pulling from library/sonarqube
e24a8b9e652f: Pull complete
4f4fb700ef54: Pull complete
f3929ce9ef98: Pull complete
1df735f481ad: Pull complete
5d5a1fad7028: Pull complete
eb27e3a98da1: Pull complete
c7ad1fe61e07: Pull complete
60d98d807669: Pull complete
ce2bd68be4fc: Download complete
600f318704dc: Download complete
Digest: sha256:f709975ab31d2d08f5a3ae2dc73a31ee011afc8cf28845082c17c55d45df9df5
Status: Downloaded newer image for sonarqube:lts-community
docker.io/library/sonarqube:lts-community
ubuntu@ip-10-0-21-61:~$ sudo docker images
                                                                                    i Info →   U In Use
IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
jenkins/jenkins:lts       a660310e39ad        720MB          246MB
nginx:latest              f9ea18bfa4fa        243MB         66.7MB
sonarqube:lts-community   f709975ab31d       1.02GB          398MB
tomcat:latest             6974cddd46f0        640MB          174MB
ubuntu:latest             f144425ff09b        162MB         45.6MB
ubuntu@ip-10-0-21-61:~$
```

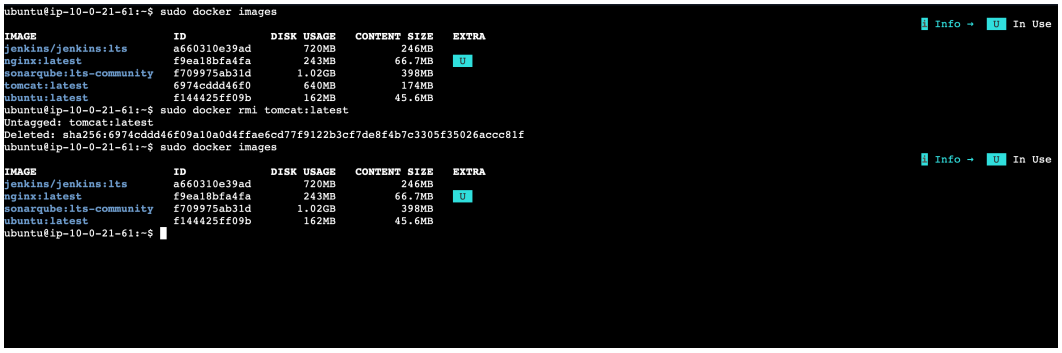
4. Run nginx container and expose on port 81.



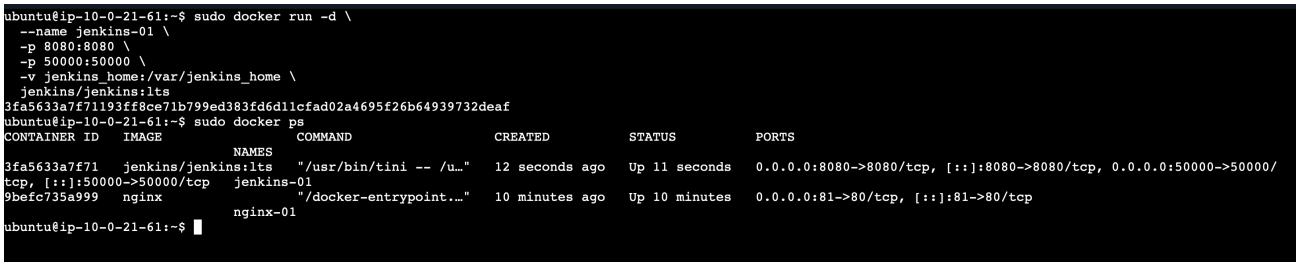
Docker-Task-1



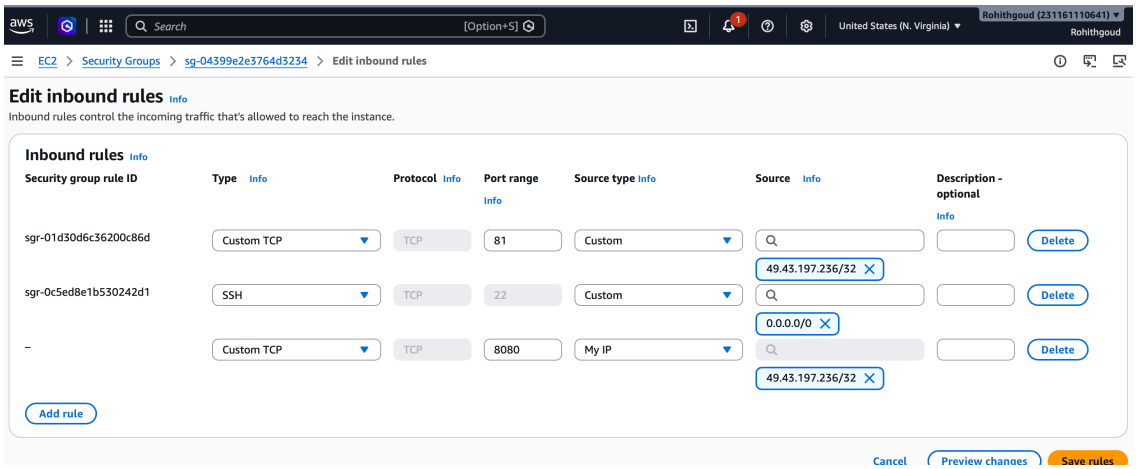
5. Delete the Apache tomcat image from local.



6. Inspect the Jenkins image, SonarQube image.

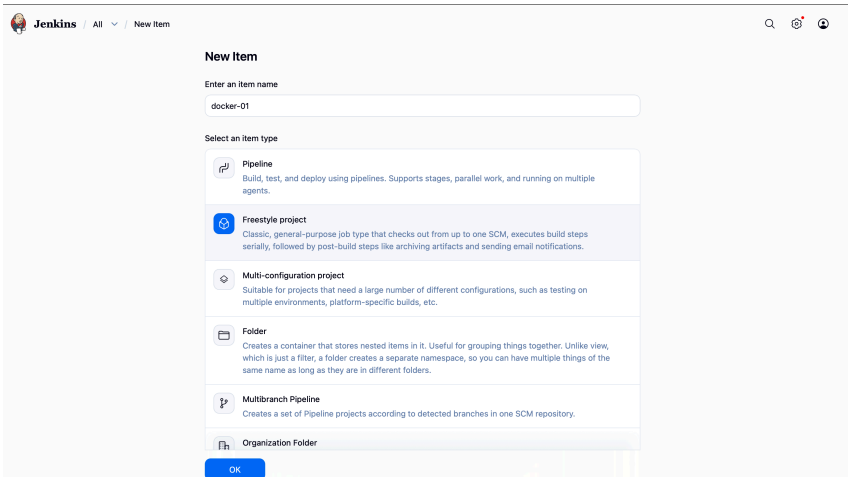


Docker-Task-1





7. Run Jenkins container and run one sample job.



Docker-Task-1

